Long-range cutoff: 6 Å vs 12 Å distance scans (PolarMACE)

This is the PolarMACE companion to `SCAN_NOTES.md` (plain MACE, 15 epochs) and `SCAN_NOTES_ep30.md` (plain MACE, 30 epochs). Two PolarMACE models were trained from scratch on the same 20k OMol25 frames as the plain MACE models (SPICE-2 elements, |q| ≤ 2, fixed ωB97M-V atom E0s, energy:force 10:10, lr 1e-3, 20 epochs). The frames were centred on their centroids (`omol25_spice20k_c`), which leaves every energy and force unchanged. The models are identical apart from the local cutoff (r_max 6 or 12 Å). Electrostatics settings are copied from the published MACE-POLAR-1 models; charge and spin enter through POLAR's charge equilibration, not embeddings.

Table 1

Model	Saved epoch	Valid RMSE_E (meV/atom)	Valid RMSE_F (meV/Å)
Plain MACE 6 Å, 30 epochs	29	85.6	226.5
Plain MACE 12 Å, 30 epochs	29	75.2	226.0
PolarMACE 6 Å (seed 7)	17	**63.5**	234.1
PolarMACE 12 Å (seed 123)	18	68.1	247.2

The scans and the method are the ones in `SCAN_NOTES.md`: eleven fragment pairs pulled apart rigidly from 2.5 to 20 Å minimum gap, ORCA ωB97M-V/def2-TZVPD at every point, ΔE(d) = E(pair) − E(A) − E(B) in kcal/mol, each curve shifted to zero at 20 Å so only the shape is compared.

Result. Explicit electrostatics fixed the ion pairs and broke the ion–neutral pairs. The cation–cation repulsion now has the right sign and nearly the right size, and the attractive ion pairs follow a 1/r tail out to 20 Å at about half the DFT strength. But every ion–neutral pair (Na^+ with water, methane and benzene; Cl^- with water) now repels with a slow 1/r-like tail, where DFT attracts. Both failures point the same way: the models spread a frame's total charge over all of its atoms, so part of an ion's charge sits on its partner. Energy-and-force training barely constrains where charge sits; the published POLAR-1-m (≈100M frames) gets all eleven pairs right. Within PolarMACE the 12 Å cutoff gives no consistent gain over 6 Å.

Summary

Shifted ΔE at 3 / 8 Å, kcal/mol.

Table 2

Pair	Interaction	DFT	POLAR 6 Å	POLAR 12 Å	12 Å vs 6 Å
NH_4^+ · acetate	attractive ion pair, r^{-1}	−76.51 / −21.51	−17.88 / −3.39	−29.11 / −8.58	12 Å better
NH_4^+ · Cl^-	attractive ion pair, r^{-1}	−74.75 / −21.22	−35.93 / −11.84	−39.67 / −12.52	Same
NH_4^+ · NH_4^+	repulsive ion pair, r^{-1}	+51.05 / +18.00	+54.60 / +23.57	+45.74 / +22.20	Same
Na^+ · water	ion–dipole, r^{-2}	−15.90 / −1.72	−11.09 / −0.88	+14.58 / +6.29	12 Å wrong sign

Cl^- · water	ion–dipole, r^{-2}	−11.46 / −1.51	+5.83 / +2.36	+5.64 / +2.35	Both wrong sign
Na^+ · methane	ion–induced dipole, r^{-4}	−3.31 / −0.08	+7.21 / +2.67	+20.12 / +6.84	Both wrong sign, 12 Å worse
Na^+ · benzene	cation–π and induction	−24.17 / −1.20	+13.45 / +6.15	+15.20 / +11.61	Both wrong sign
Water dimer, aligned dipoles	dipole–dipole, r^{-3}	−2.02 / −0.14	−2.24 / −0.14	−2.89 / −0.38	6 Å better
Acetonitrile dimer, head to tail	dipole–dipole, r^{-3}	−3.04 / −0.34	−1.78 / −0.31	−2.24 / −0.24	Same
Methane dimer	dispersion, r^{-6}	−0.23 / −0.00	−0.44 / −0.03	−1.14 / −0.27	6 Å better
Benzene dimer, stacked	dispersion and quadrupole	+11.68 / +0.01	+6.68 / −0.38	+7.75 / −2.68	6 Å better

All models against DFT

Mean absolute error of the shifted curve over all 17 gaps, kcal/mol. POLAR-1-m is the published foundation model (6 Å local cutoff), from `scan/results_polar1/`.

Table 3

Pair	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å	POLAR-1-m
NH_4^+ · NH_4^+	23.80	38.68	4.00	2.96	0.19
NH_4^+ · Cl^-	29.26	28.96	14.65	13.60	1.88
NH_4^+ · acetate	28.67	27.57	24.63	19.15	2.18
Na^+ · water	2.07	0.54	1.22	12.22	0.33
Cl^- · water	2.00	2.38	6.32	6.55	0.36
Na^+ · methane	1.14	0.98	4.13	9.75	0.37
Na^+ · benzene	3.74	2.59	13.47	17.96	0.65
Water dimer	0.20	0.17	0.08	0.31	0.01
Acetonitrile dimer	0.26	1.97	0.34	0.38	0.03
Methane dimer	0.10	0.41	0.06	0.58	0.02
Benzene dimer	1.35	4.80	2.00	4.91	1.21

Raw ΔE at 20 Å (the constant offset the shifted curves remove), kcal/mol:

Table 4

Pair	DFT	MACE 6 Å	MACE 12 Å	POLAR 6 Å	POLAR 12 Å	POLAR-1-m

Na^+ · water	−0.33	+17.74	+39.90	+18.97	−12.85	−0.65
Cl^- · water	−0.31	−3.89	−11.99	−35.95	−42.74	−0.59
Na^+ · methane	−0.00	+18.57	+46.15	+13.49	−6.24	−0.51
Na^+ · benzene	−0.07	+36.50	+77.85	+14.83	−0.27	−0.80
NH_4^+ · acetate	−15.85	−19.22	−27.48	−74.89	−73.88	−19.74
NH_4^+ · Cl^-	−15.80	−31.36	−57.06	−84.98	−97.53	−19.40
NH_4^+ · NH_4^+	+15.06	+16.59	+125.53	+21.88	+22.25	+15.15

Neutral pairs are within 0.2 kcal/mol of zero for every model.

Why the ion–neutral pairs fail

- **Wrong-sign tails.** Na^+ · benzene repels with a smooth tail that is still +1.3 (6 Å) and +2.0 (12 Å) kcal/mol at 16 Å. A neutral partner cannot produce a 1/r tail; a fraction of the cation's charge on the benzene can.
- **Half-strength ion pairs.** NH_4^+ · Cl^- follows the right 1/r shape at about half the DFT size, as if each ion carried roughly ±0.7 instead of ±1.
- **Offsets.** The raw 20 Å offsets for the ion pairs (−75 to −98 kcal/mol against DFT −16) and for Cl^- · water (−36 to −43 against −0.3) are what a wrong charge partition between separated fragments gives.
- **Training signal.** The loss had energies and forces only. Nothing directly tells the charge model where charge lives, so it can put fractional charge on any atom.
- **Next lever.** The OMol25 source frames carry Löwdin and Mulliken charges and a dipole (per cafquant's notes). A dipole or charge term in the loss should pin the partition.

Training notes

- **k-grid memory.** PolarMACE builds a k-space grid from each frame's cell even for non-periodic frames, where both electrostatics calls take the real-space path and never read it. mace sizes that cell as (max|pos|+1)·5·r_max and builds the grid for the whole batch from its largest frame. That alone asked for 55 GiB on the first launch. Centring the frames (cafquant's fix) shrinks it; the trainer and `eval_scans.py` / `frame_errors.py` also shrink the unused grid to k = 0 (`code/modal_finetune.py`).
- **6 Å instability.** The first 6 Å run (seed 123) never converged: spikes of 10–1000× in force error, also after a restart at lr 3e-4. Seed 7, same data, code and lr, converged (with transient spikes at epochs 8, 12 and 18, each recovered within one or two epochs). So the seed-123 run was a bad random draw. `frame_errors.py` on its epoch-7 state put 98% of the squared validation force error on a single 5-atom dication (HBF_2Mg, q = +2); the fragment gap was not the cause.
- **Training-set outliers.** Both final models are clean on validation but have catastrophic predictions on a few training frames (training RMSE_F 3,382 meV/Å for 12 Å and 5.1×10^6 for 6 Å). Likely the same tiny charged metal frames; check before using these models on new systems.

Plots and full tables

Ion–ion

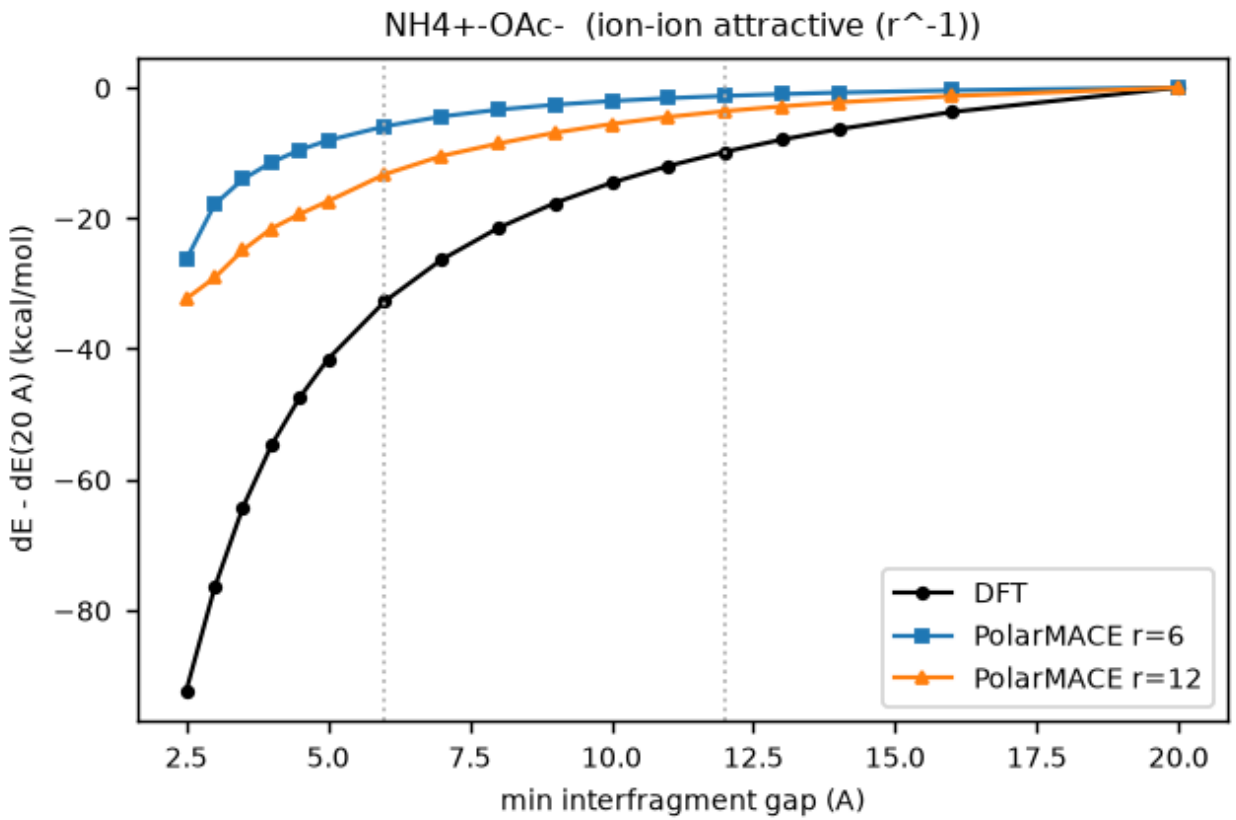


Table 5

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−92.29	−26.32	−32.32	+59.97
3.00	−76.51	−17.88	−29.11	+47.40
3.50	−64.20	−14.02	−24.86	+39.34
4.00	−54.78	−11.47	−21.60	+33.18
4.50	−47.44	−9.55	−19.38	+28.06
5.00	−41.56	−8.10	−17.40	+24.17
6.00	−32.73	−5.94	−13.28	+19.45
7.00	−26.36	−4.44	−10.48	+15.88
8.00	−21.51	−3.39	−8.58	+12.93
9.00	−17.68	−2.64	−6.92	+10.76
10.00	−14.57	−2.07	−5.59	+8.99
11.00	−12.00	−1.63	−4.50	+7.50
12.00	−9.84	−1.28	−3.62	+6.22
13.00	−7.98	−1.00	−2.89	+5.10
14.00	−6.38	−0.78	−2.27	+4.11
16.00	−3.75	−0.43	−1.30	+2.45

20.00	+0.00	+0.00	+0.00	+0.00

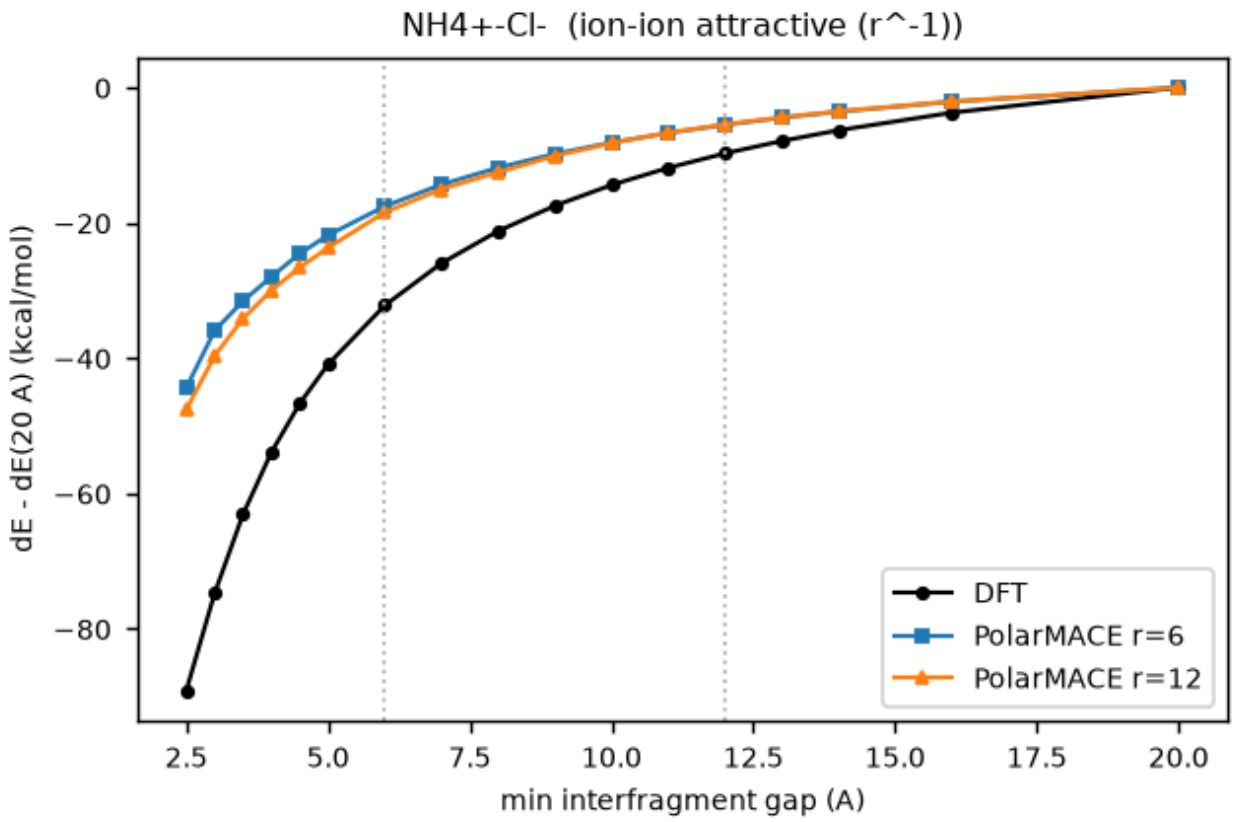


Table 6

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−89.12	−44.39	−47.60	+41.52
3.00	−74.75	−35.93	−39.67	+35.07
3.50	−63.08	−31.56	−34.15	+28.93
4.00	−53.92	−27.98	−30.00	+23.93
4.50	−46.70	−24.54	−26.61	+20.09
5.00	−40.91	−21.84	−23.68	+17.23
6.00	−32.21	−17.55	−18.46	+13.75
7.00	−25.96	−14.34	−15.02	+10.94
8.00	−21.22	−11.84	−12.52	+8.69
9.00	−17.47	−9.82	−10.12	+7.35
10.00	−14.41	−8.14	−8.22	+6.19
11.00	−11.87	−6.72	−6.70	+5.18
12.00	−9.74	−5.51	−5.45	+4.29
13.00	−7.91	−4.48	−4.41	+3.50

14.00	−6.32	−3.58	−3.51	+2.81
16.00	−3.72	−2.10	−2.05	+1.66
20.00	+0.00	+0.00	+0.00	+0.00

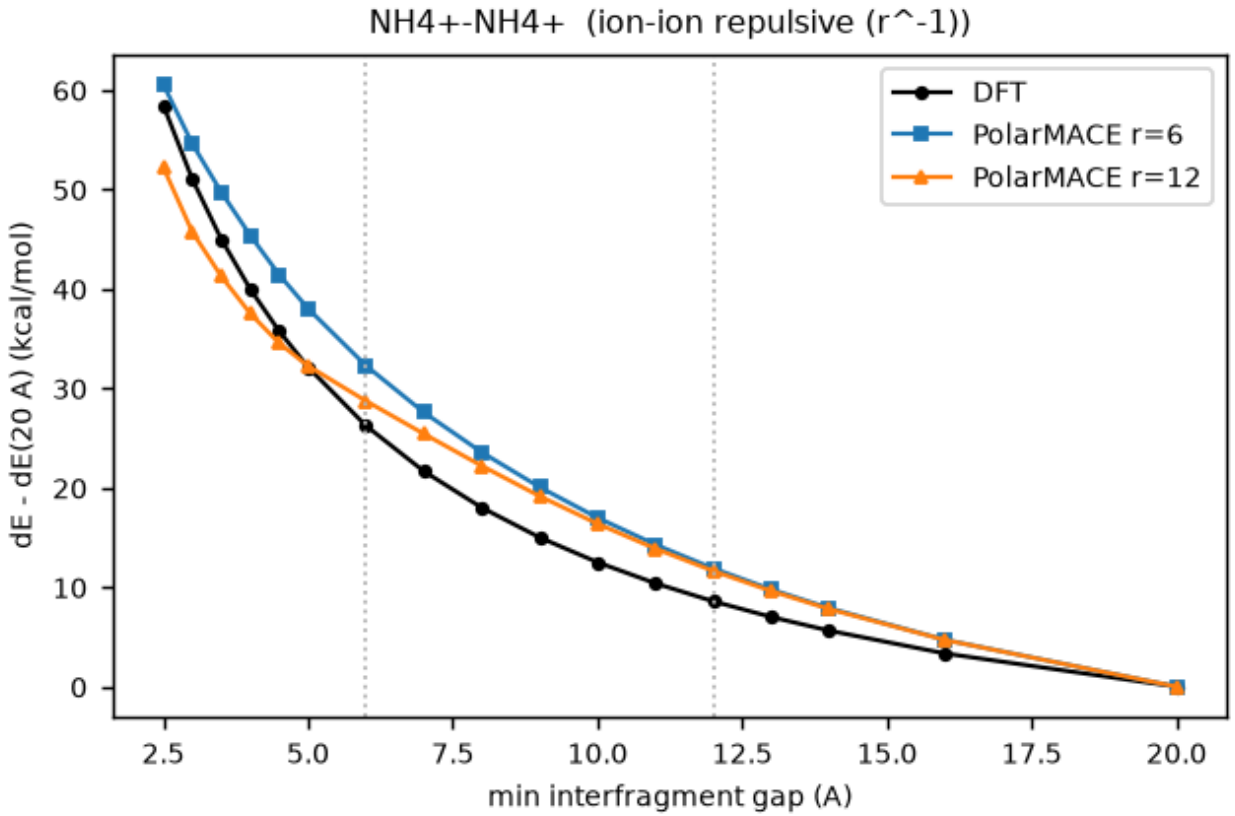


Table 7

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	+58.47	+60.57	+52.22	−6.25
3.00	+51.05	+54.60	+45.74	−5.31
3.50	+45.01	+49.74	+41.27	−3.74
4.00	+39.99	+45.40	+37.54	−2.45
4.50	+35.76	+41.43	+34.54	−1.21
5.00	+32.13	+38.02	+32.26	+0.13
6.00	+26.24	+32.28	+28.73	+2.49
7.00	+21.66	+27.58	+25.42	+3.76
8.00	+18.00	+23.57	+22.20	+4.20
9.00	+15.00	+20.06	+19.16	+4.15
10.00	+12.50	+16.99	+16.38	+3.87
11.00	+10.39	+14.28	+13.88	+3.49

12.00	+8.58	+11.89	+11.63	+3.05
13.00	+7.01	+9.77	+9.61	+2.61
14.00	+5.63	+7.89	+7.80	+2.17
16.00	+3.34	+4.71	+4.69	+1.35
20.00	+0.00	+0.00	+0.00	+0.00

Ion–neutral

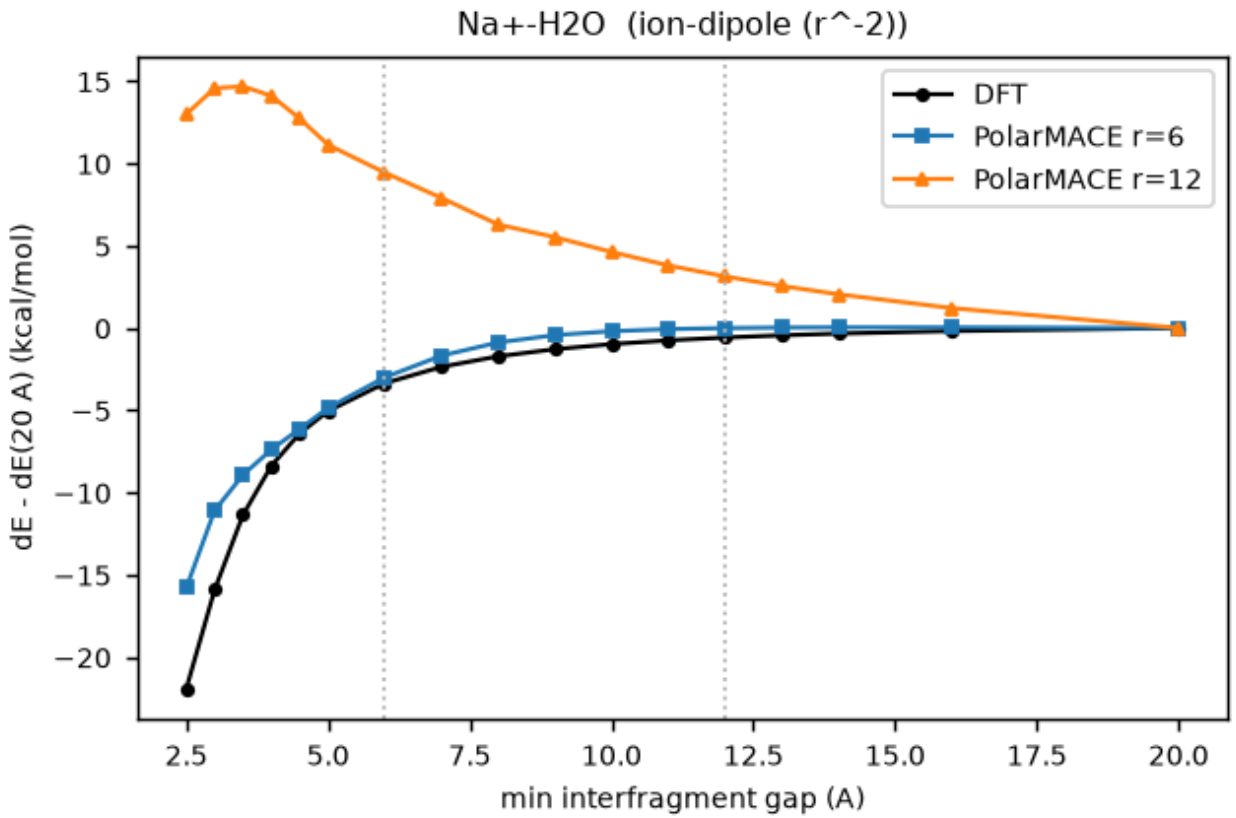


Table 8

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−21.92	−15.79	+12.99	+34.91
3.00	−15.90	−11.09	+14.58	+30.48
3.50	−11.35	−8.93	+14.68	+26.04
4.00	−8.38	−7.39	+14.11	+22.48
4.50	−6.40	−6.13	+12.73	+19.13
5.00	−5.05	−4.86	+11.13	+16.18
6.00	−3.35	−3.00	+9.44	+12.80
7.00	−2.36	−1.68	+7.90	+10.26
8.00	−1.72	−0.88	+6.29	+8.01

9.00	−1.29	−0.43	+5.52	+6.81
10.00	−0.98	−0.19	+4.62	+5.60
11.00	−0.75	−0.07	+3.80	+4.55
12.00	−0.58	−0.01	+3.14	+3.71
13.00	−0.45	+0.03	+2.56	+3.00
14.00	−0.34	+0.04	+2.05	+2.39
16.00	−0.18	+0.04	+1.21	+1.39
20.00	+0.00	+0.00	+0.00	+0.00

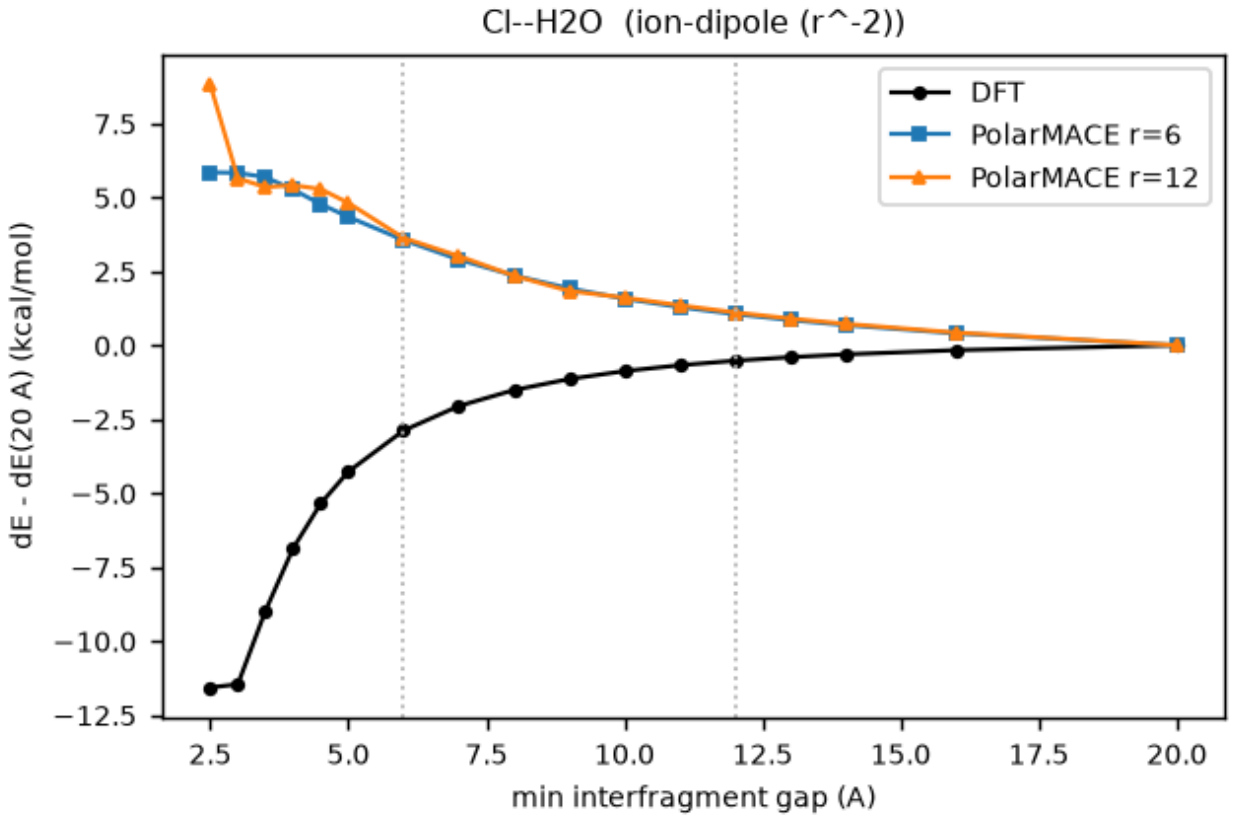


Table 9

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−11.57	+5.84	+8.81	+20.39
3.00	−11.46	+5.83	+5.64	+17.11
3.50	−8.98	+5.68	+5.35	+14.33
4.00	−6.87	+5.26	+5.41	+12.28
4.50	−5.36	+4.77	+5.28	+10.63
5.00	−4.28	+4.34	+4.81	+9.09
6.00	−2.89	+3.56	+3.63	+6.51

7.00	−2.06	+2.90	+3.01	+5.07
8.00	−1.51	+2.36	+2.35	+3.86
9.00	−1.14	+1.92	+1.83	+2.98
10.00	−0.88	+1.57	+1.61	+2.49
11.00	−0.68	+1.29	+1.35	+2.03
12.00	−0.52	+1.05	+1.11	+1.63
13.00	−0.41	+0.85	+0.90	+1.31
14.00	−0.31	+0.68	+0.72	+1.03
16.00	−0.17	+0.40	+0.43	+0.59
20.00	+0.00	+0.00	+0.00	+0.00

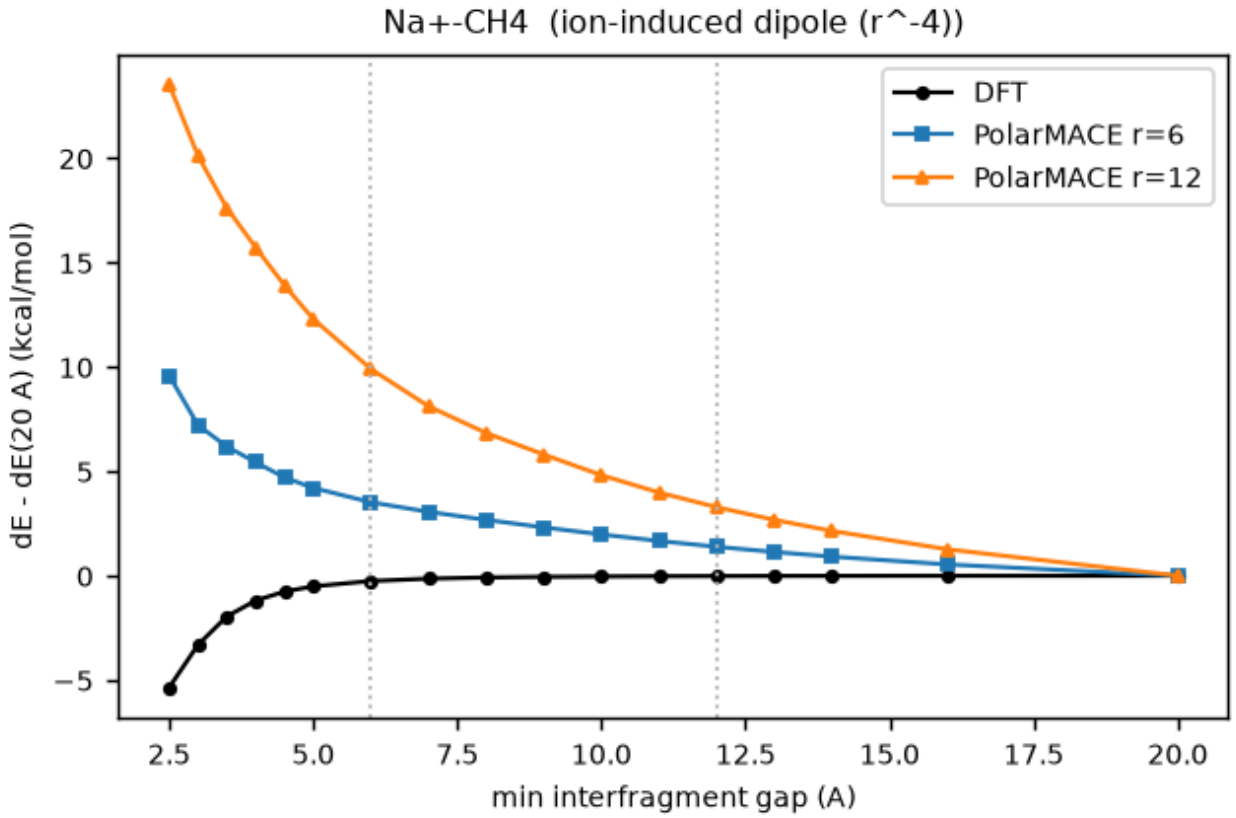


Table 10

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−5.35	+9.58	+23.53	+28.88
3.00	−3.31	+7.21	+20.12	+23.43
3.50	−1.95	+6.18	+17.61	+19.56
4.00	−1.19	+5.44	+15.72	+16.91
4.50	−0.76	+4.71	+13.90	+14.66

5.00	−0.51	+4.21	+12.30	+12.80
6.00	−0.25	+3.52	+9.90	+10.15
7.00	−0.14	+3.06	+8.12	+8.26
8.00	−0.08	+2.67	+6.84	+6.92
9.00	−0.05	+2.31	+5.80	+5.86
10.00	−0.04	+1.98	+4.81	+4.85
11.00	−0.02	+1.67	+3.98	+4.01
12.00	−0.02	+1.39	+3.28	+3.30
13.00	−0.01	+1.14	+2.67	+2.69
14.00	−0.01	+0.91	+2.15	+2.15
16.00	−0.00	+0.54	+1.26	+1.27
20.00	+0.00	+0.00	+0.00	+0.00

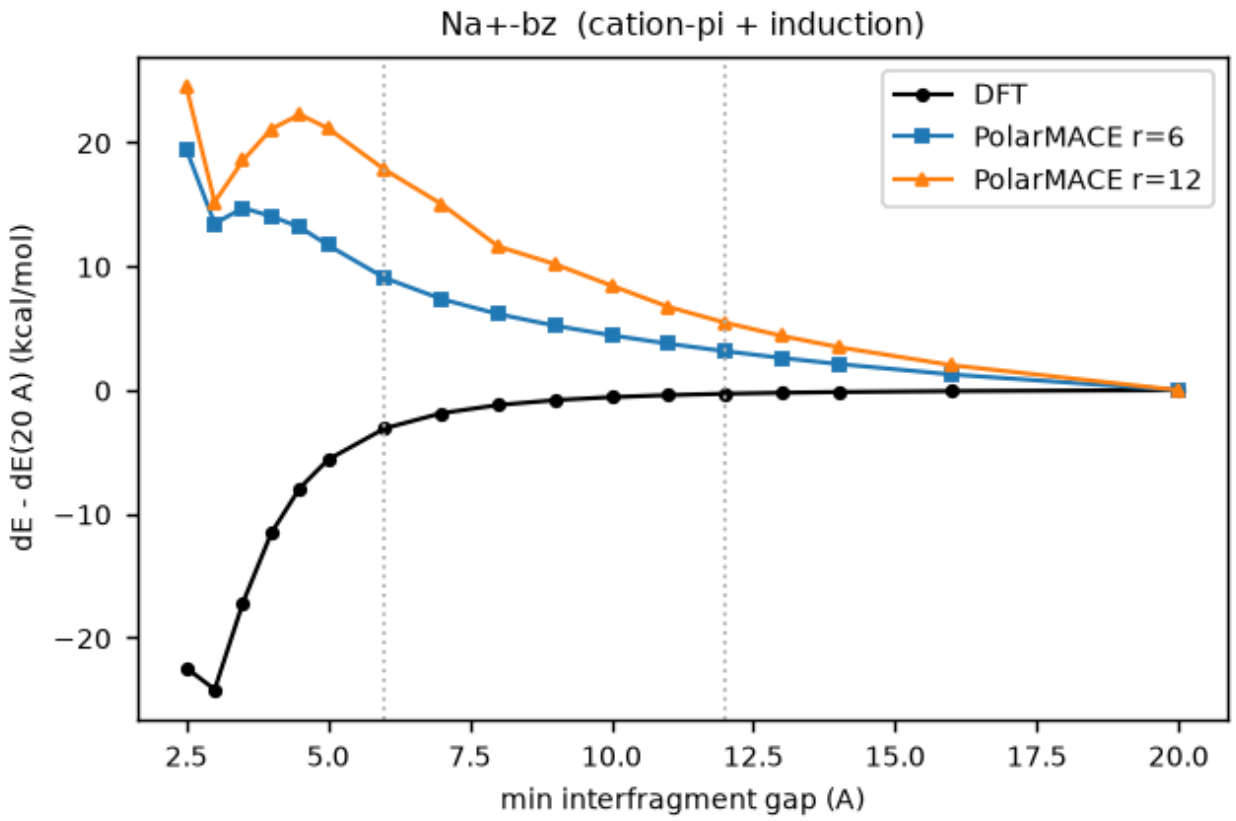


Table 11

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−22.43	+19.46	+24.52	+46.95
3.00	−24.17	+13.45	+15.20	+39.37
3.50	−17.14	+14.72	+18.56	+35.70

4.00	−11.53	+14.04	+21.09	+32.62
4.50	−7.91	+13.17	+22.26	+30.17
5.00	−5.61	+11.71	+21.15	+26.76
6.00	−3.08	+9.06	+17.80	+20.89
7.00	−1.86	+7.36	+15.03	+16.89
8.00	−1.20	+6.15	+11.61	+12.81
9.00	−0.81	+5.21	+10.18	+10.99
10.00	−0.56	+4.43	+8.44	+9.00
11.00	−0.40	+3.75	+6.73	+7.13
12.00	−0.28	+3.15	+5.46	+5.74
13.00	−0.21	+2.60	+4.39	+4.59
14.00	−0.15	+2.11	+3.48	+3.63
16.00	−0.07	+1.27	+2.02	+2.09
20.00	+0.00	+0.00	+0.00	+0.00

Neutral–neutral

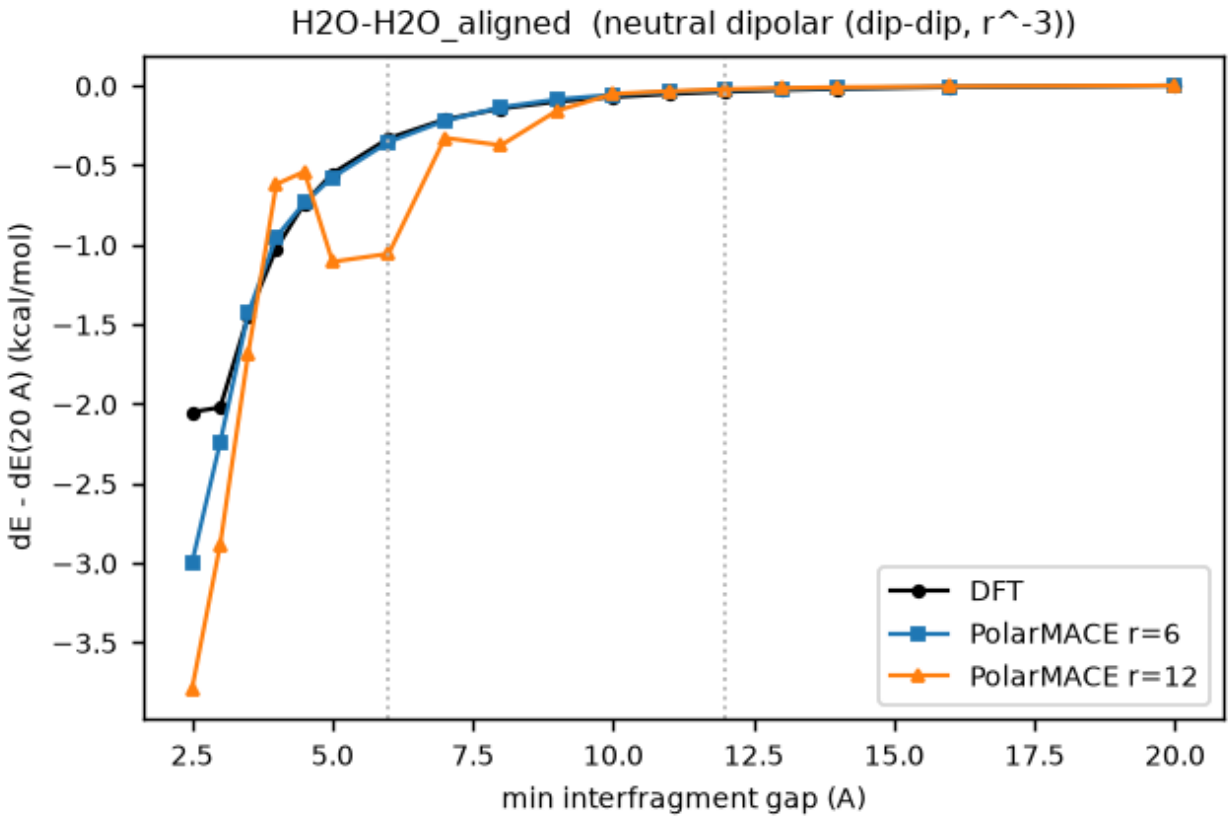


Table 12

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT

2.50	−2.05	−3.00	−3.79	−1.74
3.00	−2.02	−2.24	−2.89	−0.86
3.50	−1.45	−1.43	−1.68	−0.23
4.00	−1.02	−0.95	−0.62	+0.40
4.50	−0.74	−0.74	−0.54	+0.20
5.00	−0.55	−0.58	−1.11	−0.56
6.00	−0.33	−0.35	−1.06	−0.72
7.00	−0.21	−0.22	−0.33	−0.12
8.00	−0.14	−0.14	−0.38	−0.23
9.00	−0.10	−0.09	−0.16	−0.06
10.00	−0.07	−0.06	−0.05	+0.02
11.00	−0.05	−0.04	−0.04	+0.02
12.00	−0.04	−0.03	−0.02	+0.01
13.00	−0.03	−0.02	−0.02	+0.01
14.00	−0.02	−0.02	−0.01	+0.01
16.00	−0.01	−0.01	−0.01	+0.01
20.00	+0.00	+0.00	+0.00	+0.00

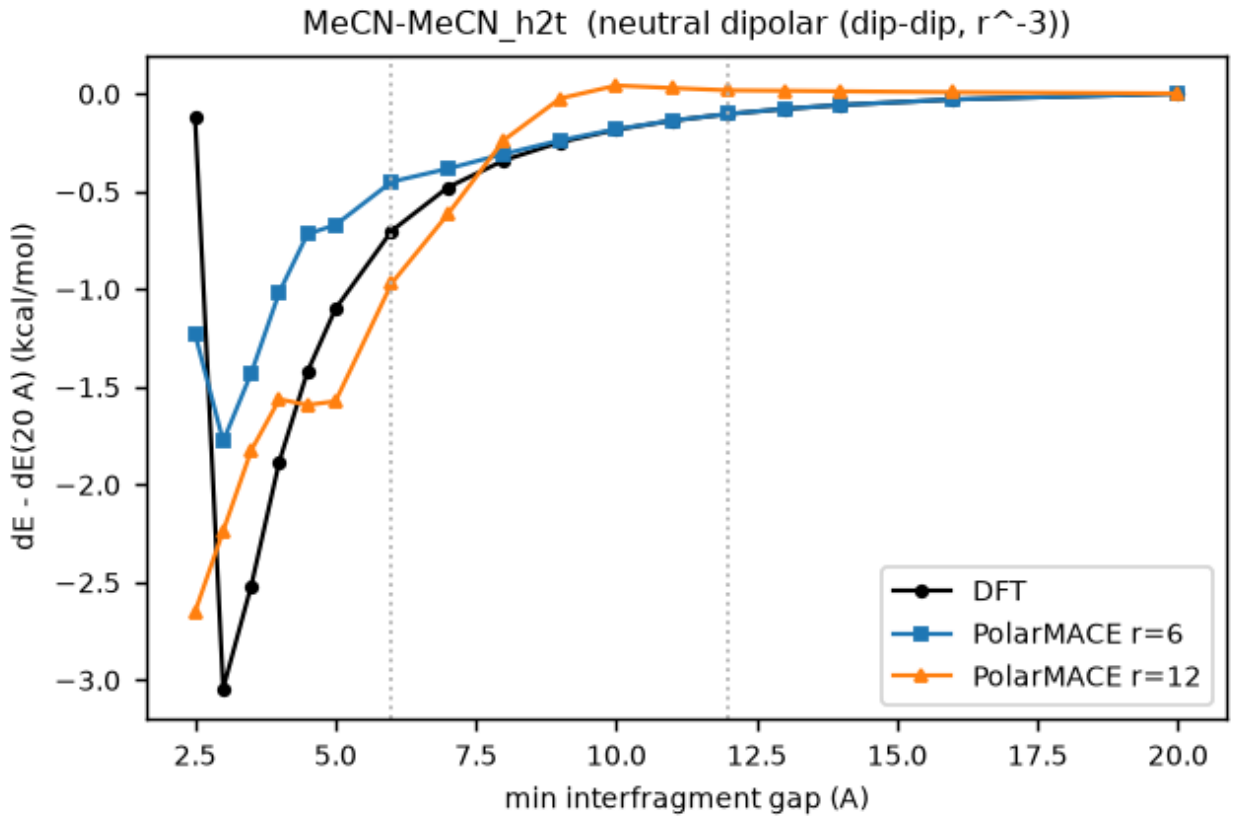


Table 13

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−0.12	−1.23	−2.65	−2.53
3.00	−3.04	−1.78	−2.24	+0.81
3.50	−2.52	−1.43	−1.83	+0.69
4.00	−1.88	−1.02	−1.56	+0.32
4.50	−1.42	−0.72	−1.59	−0.17
5.00	−1.10	−0.67	−1.57	−0.47
6.00	−0.70	−0.45	−0.97	−0.27
7.00	−0.48	−0.39	−0.62	−0.14
8.00	−0.34	−0.31	−0.24	+0.10
9.00	−0.25	−0.24	−0.03	+0.22
10.00	−0.18	−0.18	+0.04	+0.23
11.00	−0.14	−0.14	+0.03	+0.17
12.00	−0.10	−0.10	+0.02	+0.12
13.00	−0.08	−0.08	+0.01	+0.09
14.00	−0.06	−0.06	+0.01	+0.07
16.00	−0.03	−0.03	+0.01	+0.04
20.00	+0.00	+0.00	+0.00	+0.00

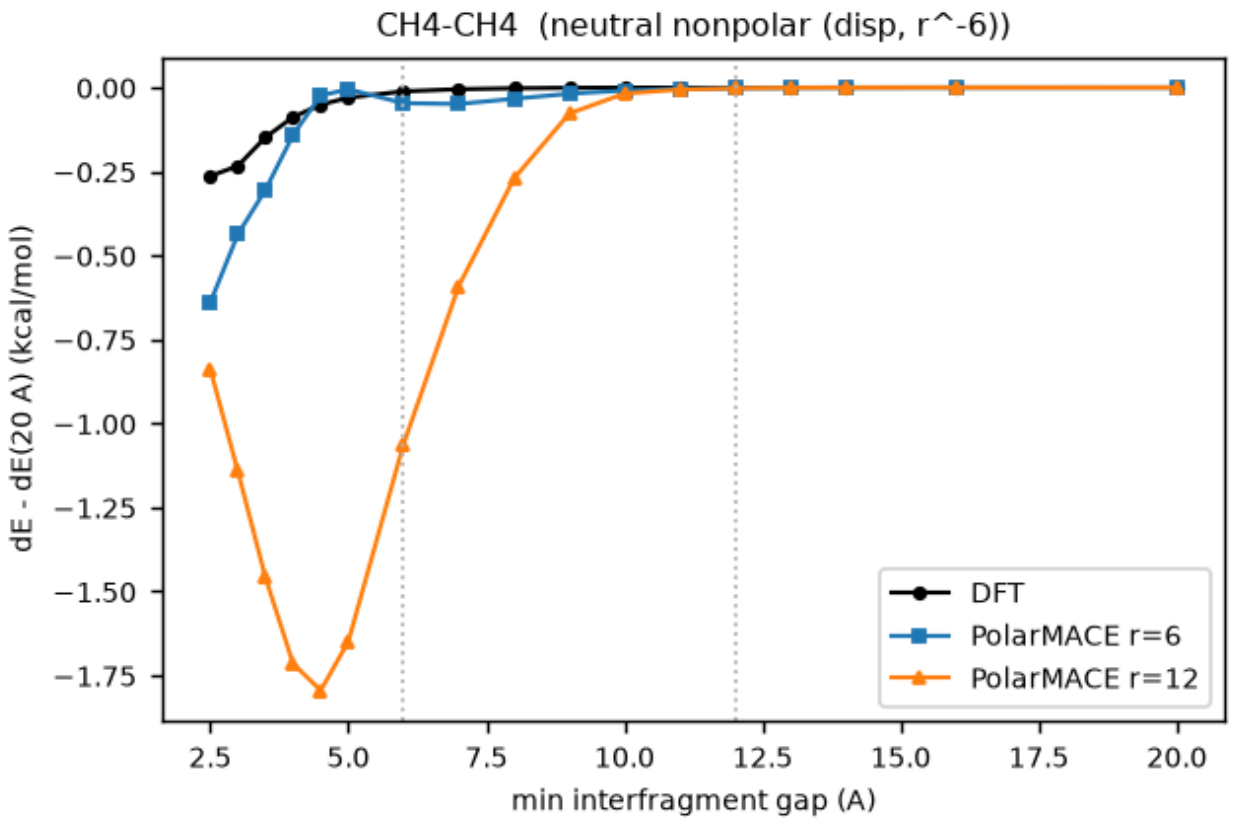


Table 14

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	−0.26	−0.64	−0.84	−0.57
3.00	−0.23	−0.44	−1.14	−0.91
3.50	−0.15	−0.30	−1.46	−1.31
4.00	−0.09	−0.14	−1.72	−1.63
4.50	−0.05	−0.02	−1.79	−1.74
5.00	−0.03	−0.01	−1.65	−1.62
6.00	−0.01	−0.05	−1.06	−1.05
7.00	−0.01	−0.05	−0.59	−0.58
8.00	−0.00	−0.03	−0.27	−0.27
9.00	−0.00	−0.02	−0.08	−0.08
10.00	−0.00	−0.01	−0.02	−0.02
11.00	−0.00	−0.00	−0.01	−0.01
12.00	−0.00	−0.00	−0.00	−0.00
13.00	−0.00	−0.00	−0.00	−0.00
14.00	−0.00	−0.00	−0.00	+0.00
16.00	−0.00	+0.00	+0.00	+0.00

20.00	+0.00	+0.00	+0.00	+0.00

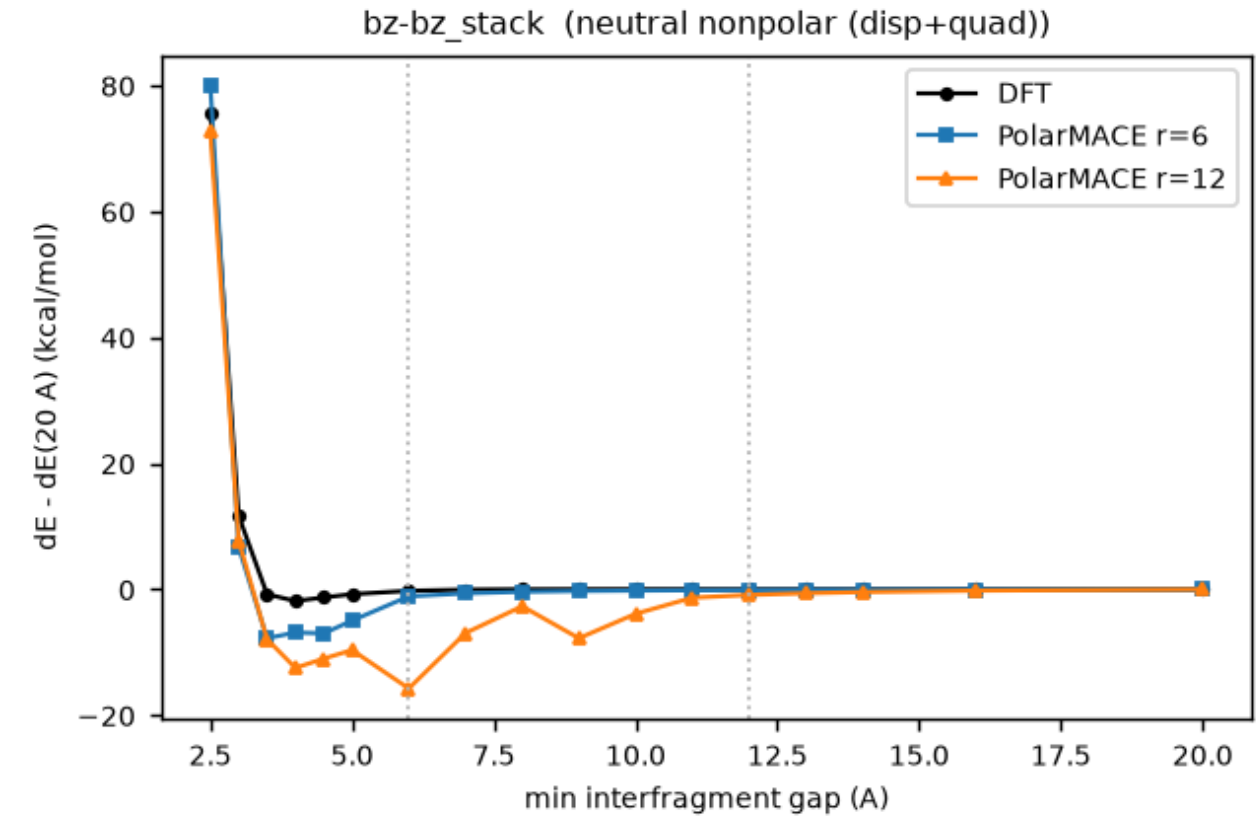


Table 15

gap (Å)	DFT	POLAR 6 Å	POLAR 12 Å	12 Å − DFT
2.50	+75.66	+80.01	+72.79	−2.86
3.00	+11.68	+6.68	+7.75	−3.93
3.50	−0.76	−7.76	−7.92	−7.15
4.00	−1.86	−6.85	−12.47	−10.61
4.50	−1.28	−6.99	−11.00	−9.72
5.00	−0.77	−4.96	−9.64	−8.87
6.00	−0.23	−1.11	−15.74	−15.51
7.00	−0.05	−0.63	−6.94	−6.89
8.00	+0.01	−0.38	−2.68	−2.69
9.00	+0.02	−0.26	−7.76	−7.78
10.00	+0.02	−0.19	−3.88	−3.91
11.00	+0.02	−0.14	−1.32	−1.34
12.00	+0.01	−0.11	−0.88	−0.90
13.00	+0.01	−0.09	−0.61	−0.62

14.00	+0.01	−0.07	−0.43	−0.43
16.00	+0.00	−0.04	−0.20	−0.20
20.00	+0.00	+0.00	+0.00	+0.00

Data: `longrange/scan/results_polar/scan_table.tsv`, from `eval_scans.py polar` (run with cafquant's `.venv/bin/python`, which has `graph_longrange`). Models: `runs/scratch/omol_spice20k_polar_r6_s7/` and `runs/scratch/omol_spice20k_polar_r12/` (`*.cpu.model`). Per-epoch history: `longrange/epoch_table.py --polar`.